Part 1a: Data Wrangling

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

#load data and inspection
data=pd.read_csv('/content/country_wise_latest.csv')

#Initial Observations
data.head()
data.info()
data.describe()

#Data Cleaning
data.dropna(inplace=True)

#Data with WHO region aggregation
data_region = data.groupby('WHO Region').agg({
    'Confirmed': 'sum',
    'Deaths': 'sum',
    'Recovered': 'sum',
    'Active': 'sum',
    'New cases': 'sum',
    'New deaths': 'sum',
    'New recovered': 'sum'
}).reset_index()

# Calculate rates after aggregation
data_region['Case_Fatality_Rate'] = (data_region['Deaths'] / data_region['Confirmed'] * 100).round(2)
data_region['Recovery_Rate'] = (data_region['Recovered'] / data_region['Confirmed'] * 100).round(2)
data_region.head()

#Save to csv
data_region.to_csv('data_region.csv', index=False)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 187 entries, 0 to 186
Data columns (total 15 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Country/Region         187 non-null    object
 1   Confirmed              187 non-null    int64
 2   Deaths                 187 non-null    int64
 3   Recovered              187 non-null    int64
 4   Active                 187 non-null    int64
 5   New cases              187 non-null    int64
 6   New deaths             187 non-null    int64
 7   New recovered          187 non-null    int64
 8   Deaths / 100 Cases     187 non-null    float64
 9   Recovered / 100 Cases  187 non-null    float64
 10  Deaths / 100 Recovered 187 non-null    float64
 11  Confirmed last week    187 non-null    int64
 12  1 week change          187 non-null    int64
 13  1 week % increase      187 non-null    float64
 14  WHO Region             187 non-null    object
dtypes: float64(4), int64(9), object(2)
memory usage: 22.0+ KB
```

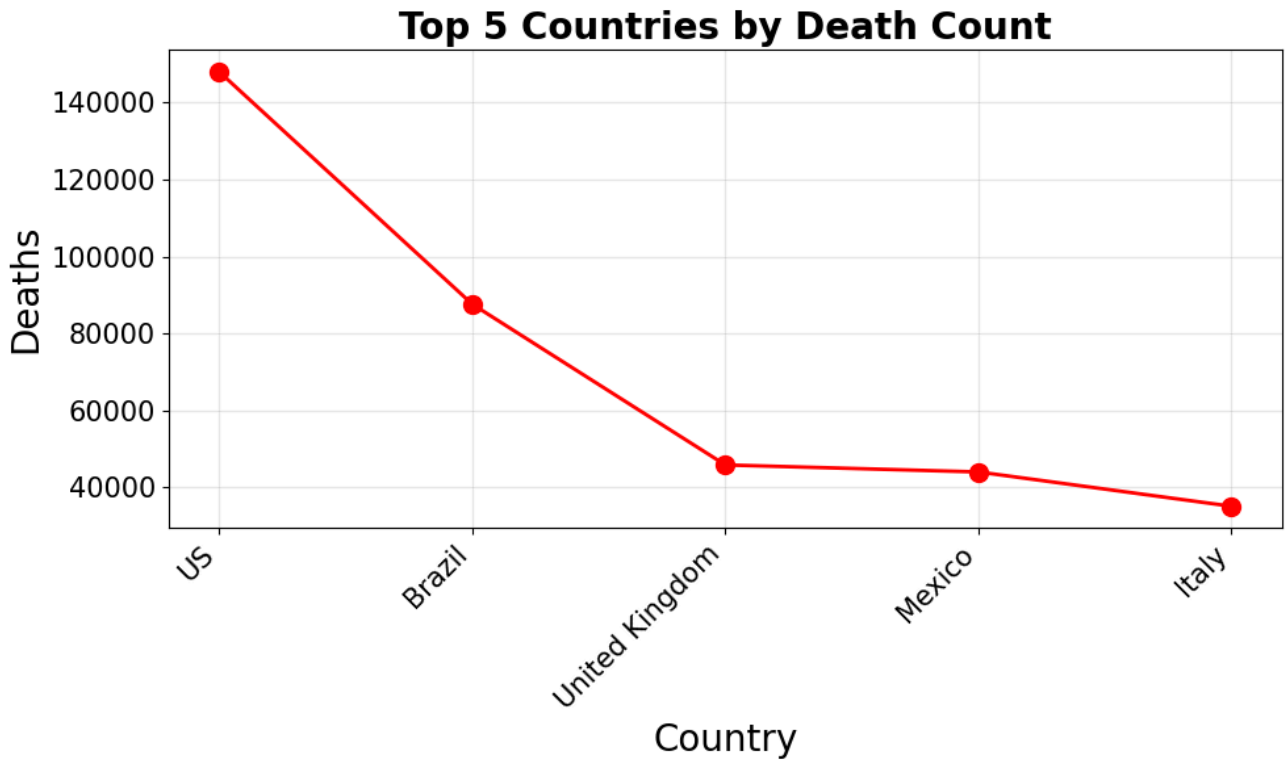
```
# Data without WHO aggregation

#Case Fatality Rate (CFR)
data['Case_Fatality_Rate']=(data['Deaths']/data['Confirmed']*100).round(2)
data.head()
```

```
#Recovery Rate
data['Recovery_Rate']=(data['Recovered']/data['Confirmed']*100).round(2)
data.head()


#Selecting Relevant columns
data=data[['Country/Region','Confirmed','Deaths','Recovered','Active','Case_Fatality_Rate','Recovery_Rate',
data.head()

#save to csv
data.to_csv('data_without_region_agg.csv',index=False)
```

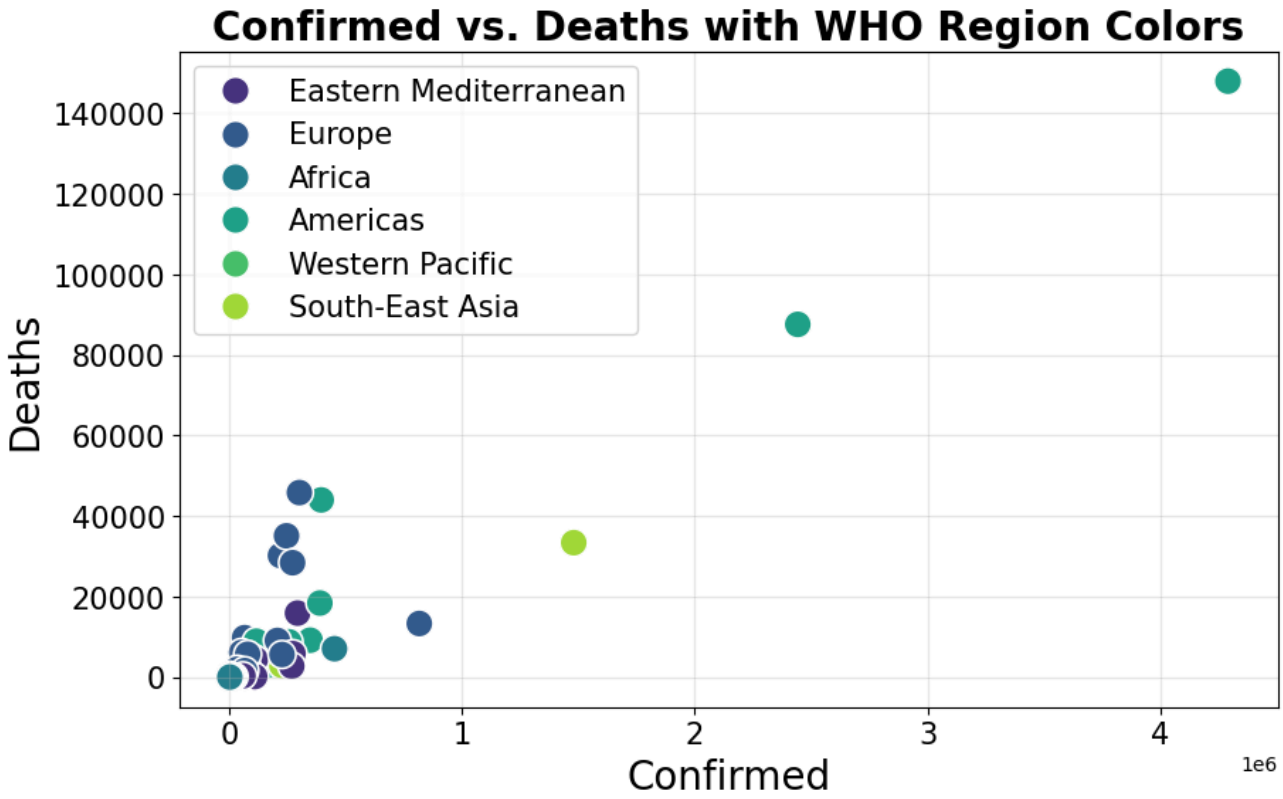
Part 2: Data Visualization with Matplotlib and Seaborn

```
#Line chart: Top 5 countries deaths over confirmed cases (hint: use Pandas nlargest)
top_deaths=data.nlargest(5,'Deaths')
print(top_deaths)
plt.figure(figsize=(10, 6))
plt.plot(top_deaths['Country/Region'], top_deaths['Deaths'],
         marker='o', linewidth=2, markersize=10, color='red')
plt.xlabel('Country', fontsize=20)
plt.ylabel('Deaths', fontsize=20)
plt.title('Top 5 Countries by Death Count', fontsize=20, fontweight='bold')
plt.xticks(rotation=45, ha='right',fontsize=15)
plt.yticks(fontsize=15)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

```
     Country/Region  Confirmed  Deaths  Recovered   Active  \
173              US    4290259  148011    1325804  2816444
23           Brazil    2442375   87618    1846641   508116
177  United Kingdom     301708   45844       1437   254427
111          Mexico     395489   44022     303810    47657
85            Italy     246286   35112     198593    12581

     Case_Fatality_Rate  Recovery_Rate  New deaths  New cases  New recovered  \
173                3.45          30.90        1076      56336          27941
23                 3.59          75.61         614      23284          33728
177               15.19           0.48           7        688              3
111               11.13          76.82         342       4973           8588
85                14.26          80.64           5        168            147

    WHO Region
173   Americas
23    Americas
177     Europe
111   Americas
85      Europe
```

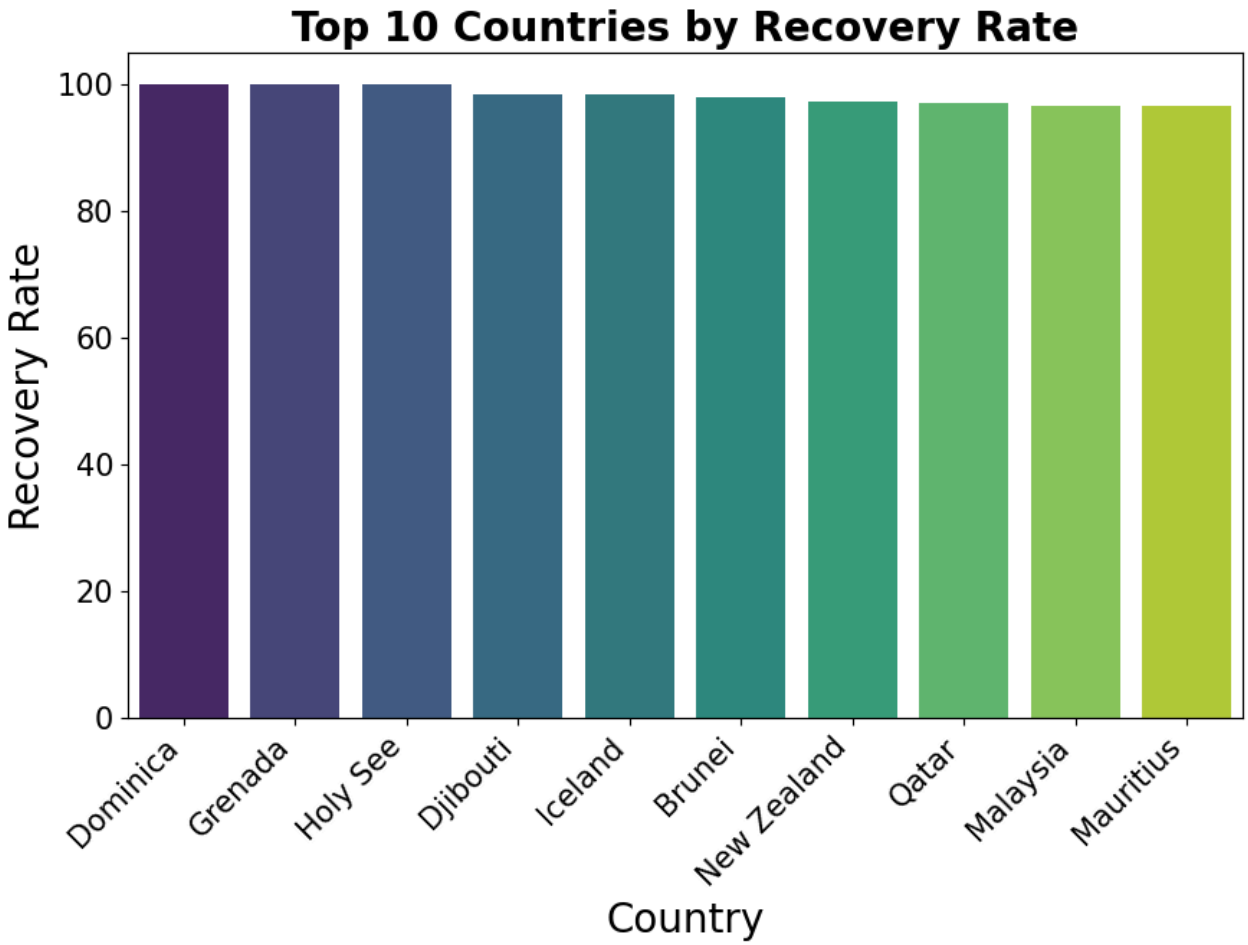



```
#Scatter plot: Confirmed vs. Deaths with WHO Region colors

plt.figure(figsize=(10, 6))
sns.scatterplot(data=data, x='Confirmed', y='Deaths', hue='WHO Region', palette='viridis',s=200)
plt.xlabel('Confirmed', fontsize=20)
plt.ylabel('Deaths', fontsize=20)
plt.title('Confirmed vs. Deaths with WHO Region Colors', fontsize=20, fontweight='bold')
plt.xticks(fontsize=15)
plt.yticks(fontsize=15)
plt.legend(fontsize=15)
plt.grid(True, alpha=0.3)
```



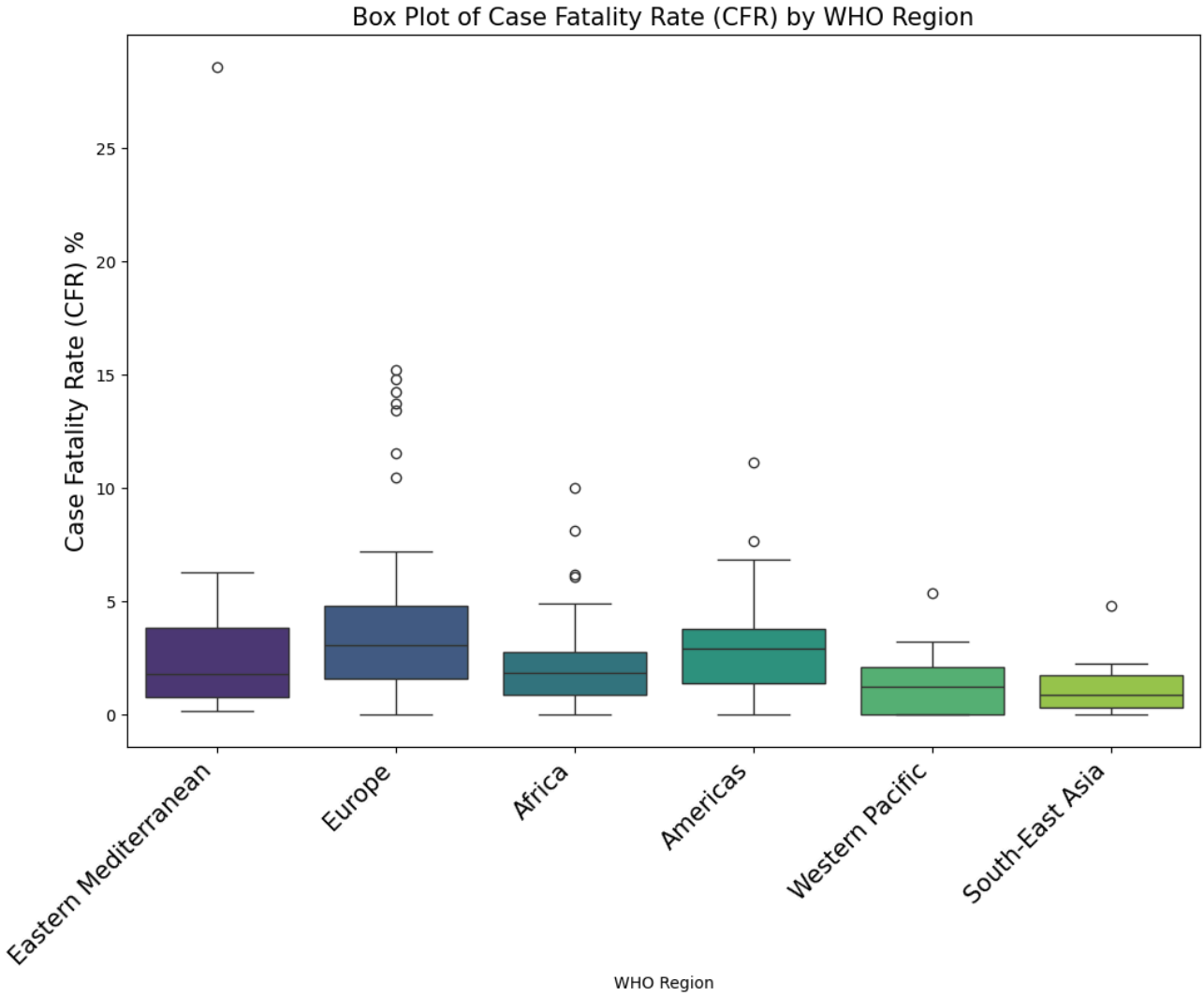

```
#Bar chart: Top 10 countries by recovery rate (hint: use Pandas nlargest)
top_recovery=data.nlargest(10,'Recovery_Rate')
plt.figure(figsize=(10, 6))
sns.barplot(data=top_recovery, x='Country/Region', y='Recovery_Rate', palette='viridis',hue='Country/Regio
plt.xlabel('Country', fontsize=20)
plt.ylabel('Recovery Rate', fontsize=20)
plt.xticks(rotation=45, ha='right',fontsize=15)
plt.yticks(fontsize=15)
plt.title('Top 10 Countries by Recovery Rate', fontsize=20, fontweight='bold')
```

```
Text(0.5, 1.0, 'Top 10 Countries by Recovery Rate')
```

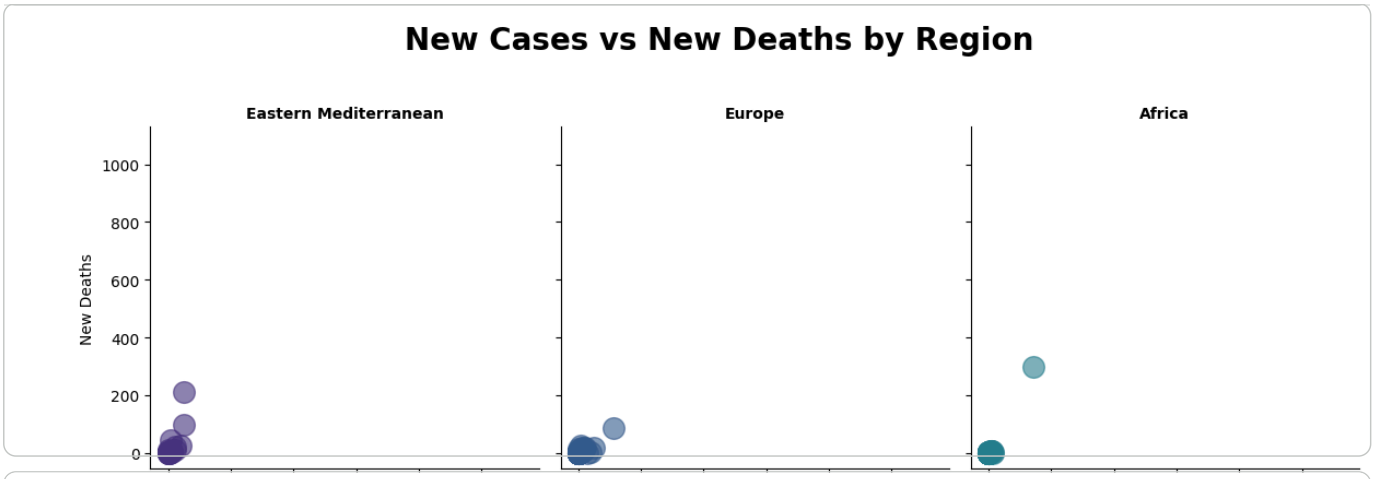



```
#Box plot: Deaths distribution by WHO Region
plt.figure(figsize=(12, 8))
sns.boxplot(x='WHO Region', y='Case_Fatality_Rate', data=data,palette='viridis',hue='WHO Region')
plt.title('Box Plot of Case Fatality Rate (CFR) by WHO Region',fontsize=15)
plt.ylabel('Case Fatality Rate (CFR) %',fontsize=15)
plt.xlabel('WHO Region')
plt.xticks(rotation=45, ha='right',fontsize=15)
plt
```

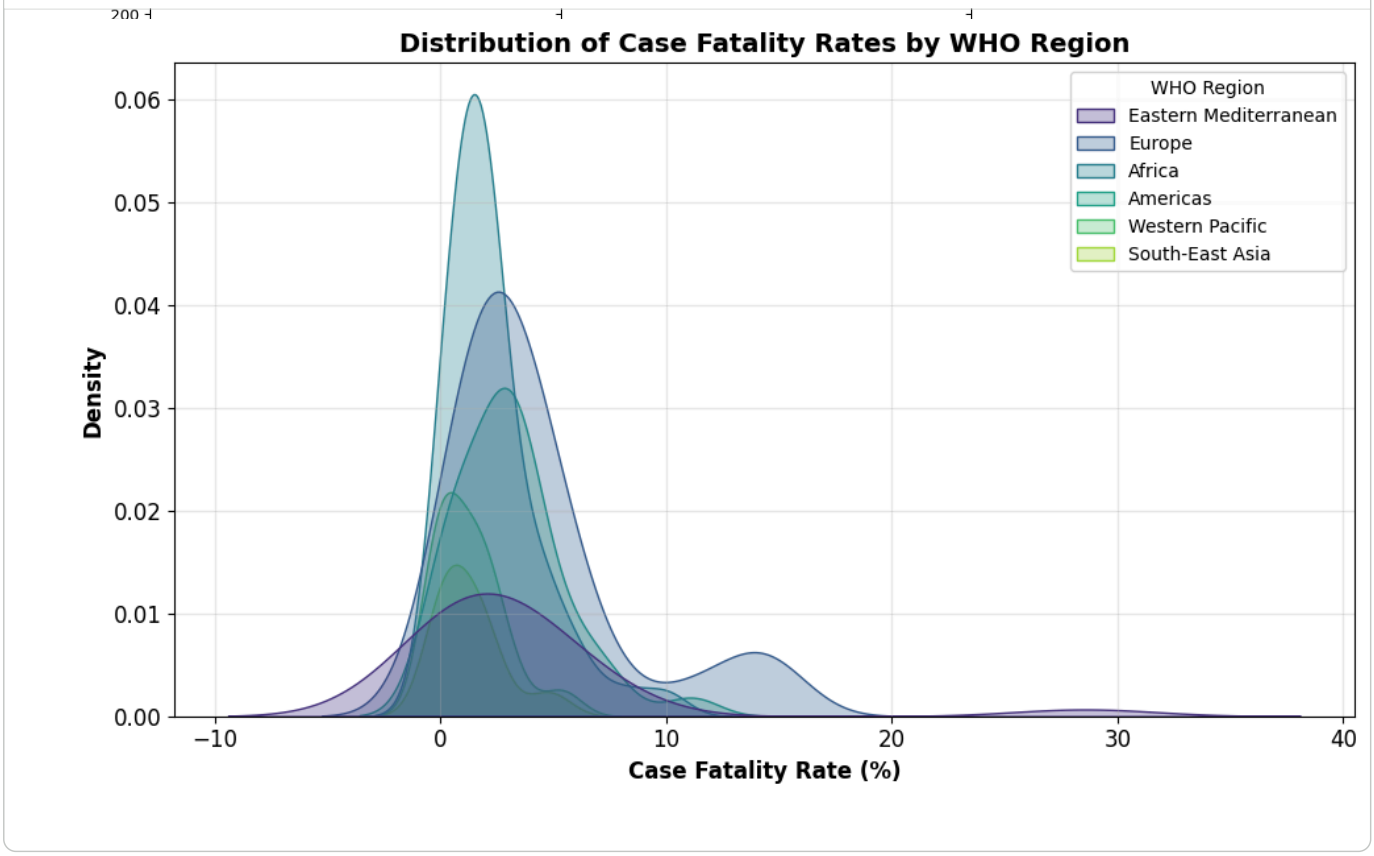
```
<module 'matplotlib.pyplot' from '/usr/local/lib/python3.12/dist-packages/matplotlib/pyplot.py'>
```




```
#Faceted plot: New cases vs New deaths by region
p = sns.FacetGrid(data=data, col='WHO Region', col_wrap=3, height=4,palette="viridis",hue='WHO Region')
p.fig.suptitle('New Cases vs New Deaths by Region', fontsize=20, fontweight='bold',y=1.02)
p.map(plt.scatter, 'New cases', 'New deaths', alpha=0.6,s=200)
p.set_titles("{col_name}", fontsize=12, fontweight='bold')
p.set_axis_labels('New Cases', 'New Deaths')
p.tight_layout()
plt.show()
```




```
plt.figure(figsize=(10, 6))
sns.kdeplot(data=data, x='Case_Fatality_Rate', hue='WHO Region', fill=True, alpha=0.3,palette="viridis")
plt.xlabel('Case Fatality Rate (%)', fontsize=12, fontweight='bold')
plt.ylabel('Density', fontsize=12, fontweight='bold')
plt.title('Distribution of Case Fatality Rates by WHO Region', fontsize=14, fontweight='bold')
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```



Part 3: Interactive Visualizations with Plotly